



Cricket Player Performance Prediction System

Leveraging Machine Learning for Predictive Analytics in Cricket

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The Challenge: Cricket Analysis

Cricket performance prediction is:

- COMPLEX
- OFTEN MANUAL
- TIME-CONSUMING
- PRONE TO SUBJECTIVE INTERPRETATION,
- LACKING THE PRECISION REQUIRED DECISION MAKING.



Objectives: Precision & Insight



Predict

Accurately forecast batsman runs and bowler wickets for upcoming matches.



Leverage Historical Data

Utilize comprehensive IPL data from 2008–2025 for robust model training.



Develop Accurate Models

Build high-performing machine learning models tailored to cricket dynamics.



Interactive Dashboard

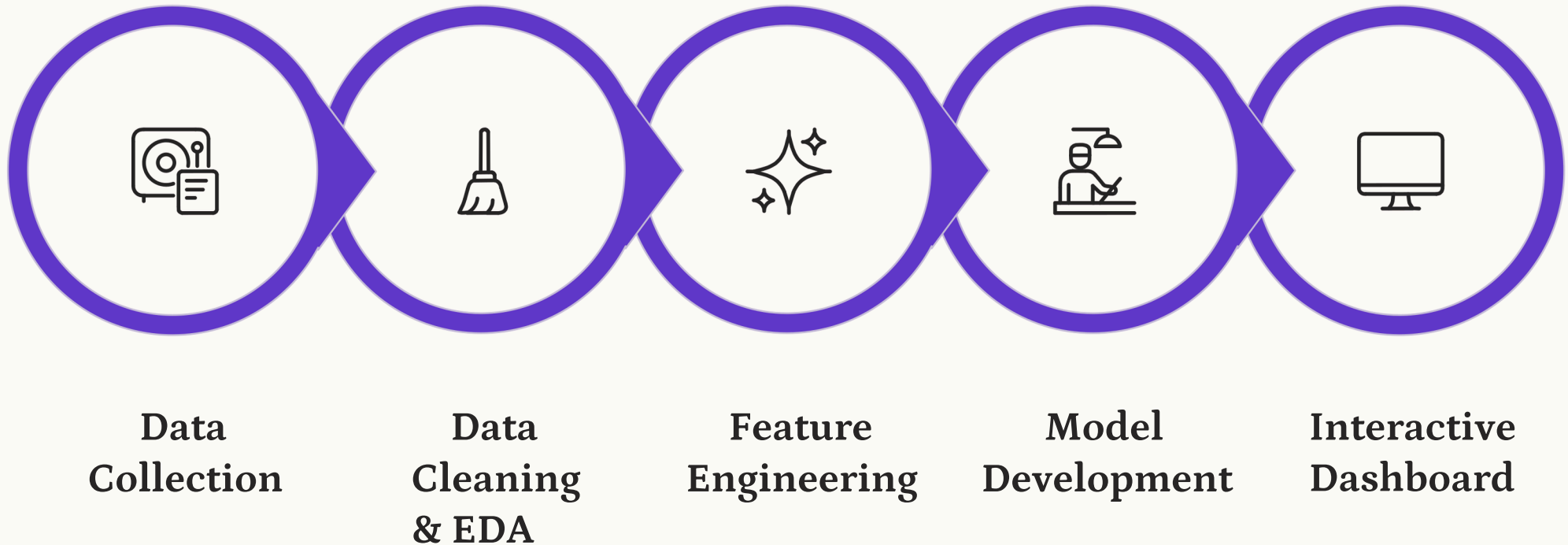
Create a user-friendly dashboard for real-time predictions and analysis.



Easy Prediction

Provide transparent, explainable predictions to foster trust and understanding.

System Design



Our system follows a structured pipeline, from raw data ingestion to an interactive predictive dashboard.

Dataset Overview: The Fuel for Our Models



Source: Kaggle, a rich repository of cricket statistics.

Dataset	Rows	Columns
ball_by_ball_data	278,205	30
ipl_matches_data	1,169	23
players-data-upda	772	6
team_aliases	46	3
teams_data	16	2

Data Refinement & Exploratory Insights

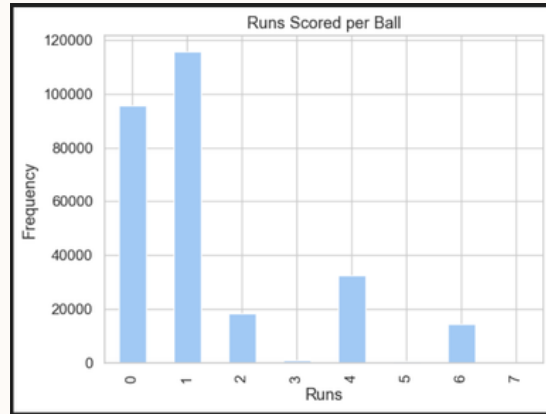
Data Cleaning:

- **Removed duplicates:** Eliminated redundant entries for accuracy.
- **Handled missing values:** Imputed or removed incomplete records strategically.
- **Standardized names:** Ensured consistent team and player nomenclature.
- **Converted formats:** Transformed dates and numeric fields for model compatibility.

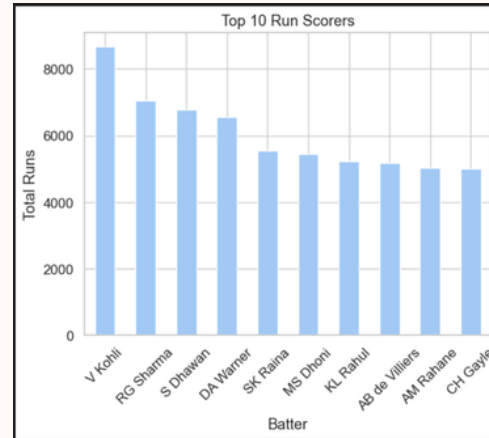


Exploratory Data Analysis (EDA): Uncovering Patterns

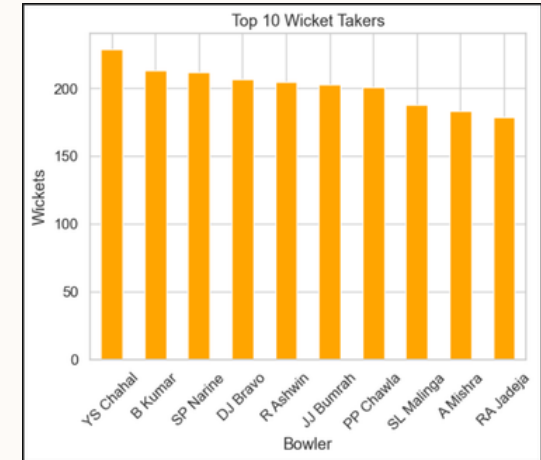
Through EDA, we gained crucial insights into the dataset's underlying structure and trends.



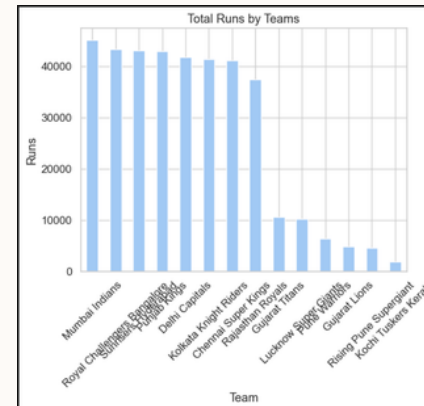
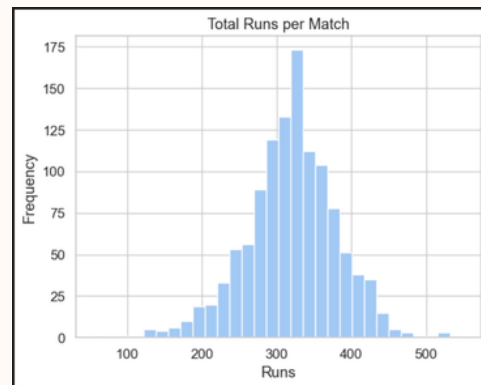
Runs Distribution



Top Runs



Top Wicket Takers



Feature Engineering: Crafting Predictive Power

The quality of our predictions hinges on carefully engineered features, transforming raw data into meaningful predictors.

1

Career Statistics

Aggregated historical performance data (total runs, wickets, averages, etc.).

2

Recent Form

Calculated performance metrics from the last 5 and 10 matches to capture current momentum.

3

Opponent-Based Features

Incorporated player performance against specific opposing teams and individuals.

4

Venue-Based Features

Accounted for how players perform at different stadiums, considering pitch conditions and home- ground advantage.



Model Selection & Training

We selected robust ensemble methods known for their predictive power and ability to handle complex datasets:

Random Forest

- **XGBoost**
- **LightGBM**

Our training process focused on

regression tasks to predict

- continuous values (runs and wickets):

Hyperparameter Tuning:

- Exhaustively optimized model using techniques like Grid Search and Random Search.

Took 12 minutes run time

Data Splitting Strategy

the dataset was divided into:

- Training Set – 80%
- Testing Set – 20%

◆ Why 80–20 Split?

- 80% data is used to train the model
- 20% data is used to test model performance
- Prevents overfitting
- Ensures unbiased evaluation

Model Evaluation

Key Evaluation Metrics

• We rigorously assessed our models using standard regression metrics:

- **Mean Absolute Error (MAE):**
- **Root Mean Squared Error (RMSE):**
- **R² Score:**

For batsman: XGB MAE: 9.045175558281114

BEST XGB RMSE: 12.169984300340385

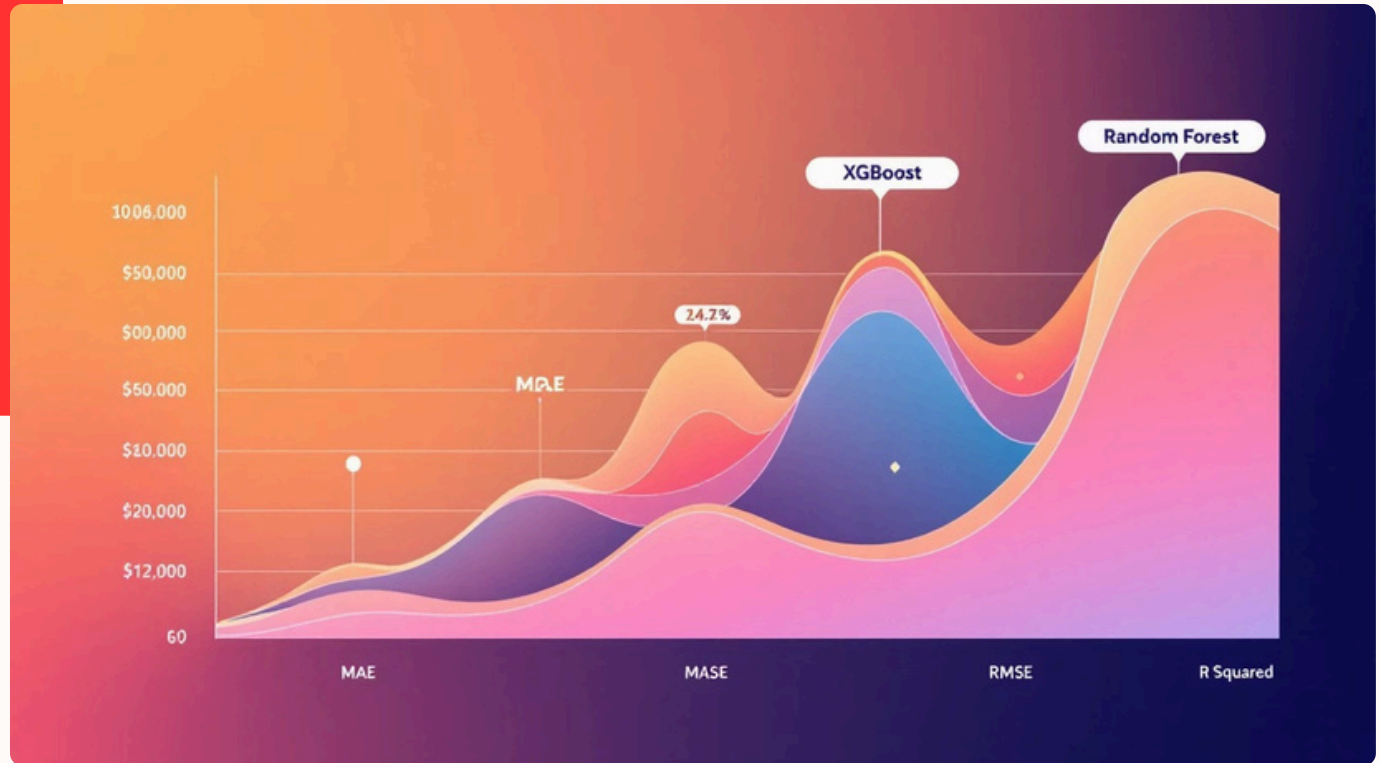
BEST XGB R2: 0.7053843337317794

For bowler:

Tuned RF MAE: 0.8295097084031978

Tuned RF RMSE: 1.05548181481273

Tuned RF R2: 0.04568238885329012





Best Model Selection:

1

For Batsman Prediction:

XGBoost consistently outperformed others, demonstrating superior accuracy in forecasting runs.

2

For Bowler Prediction:

Random Forest proved most effective in predicting wickets, offering robust and stable performance.

variables, showing model fit.

Interactive Dashboard Features



Player & Team Inputs

Seamless selection for specific players or teams.



Predicted Runs & Wickets

Clear display of forecasted performance metrics.



Last 10 Match Graphs

Visual trend analysis of recent player performance.



SHAP Explanation

Transparent insights into what drives each prediction.

Match Inputs

Select Batman
None

Select Bowler
None

Batting Team
Delhi Capitals

Bowling Team
Chennai Super Kings

Venue
Arun Jaitley Stadium

Predict Performance

Cricket Player Performance Prediction

Predicted Runs
149

Predicted Wickets
6

Match Summary

	Team	Avg Runs Per Match	Avg Wickets Per Match
0	Delhi Capitals	149	6
1	Chennai Super Kings	155	5

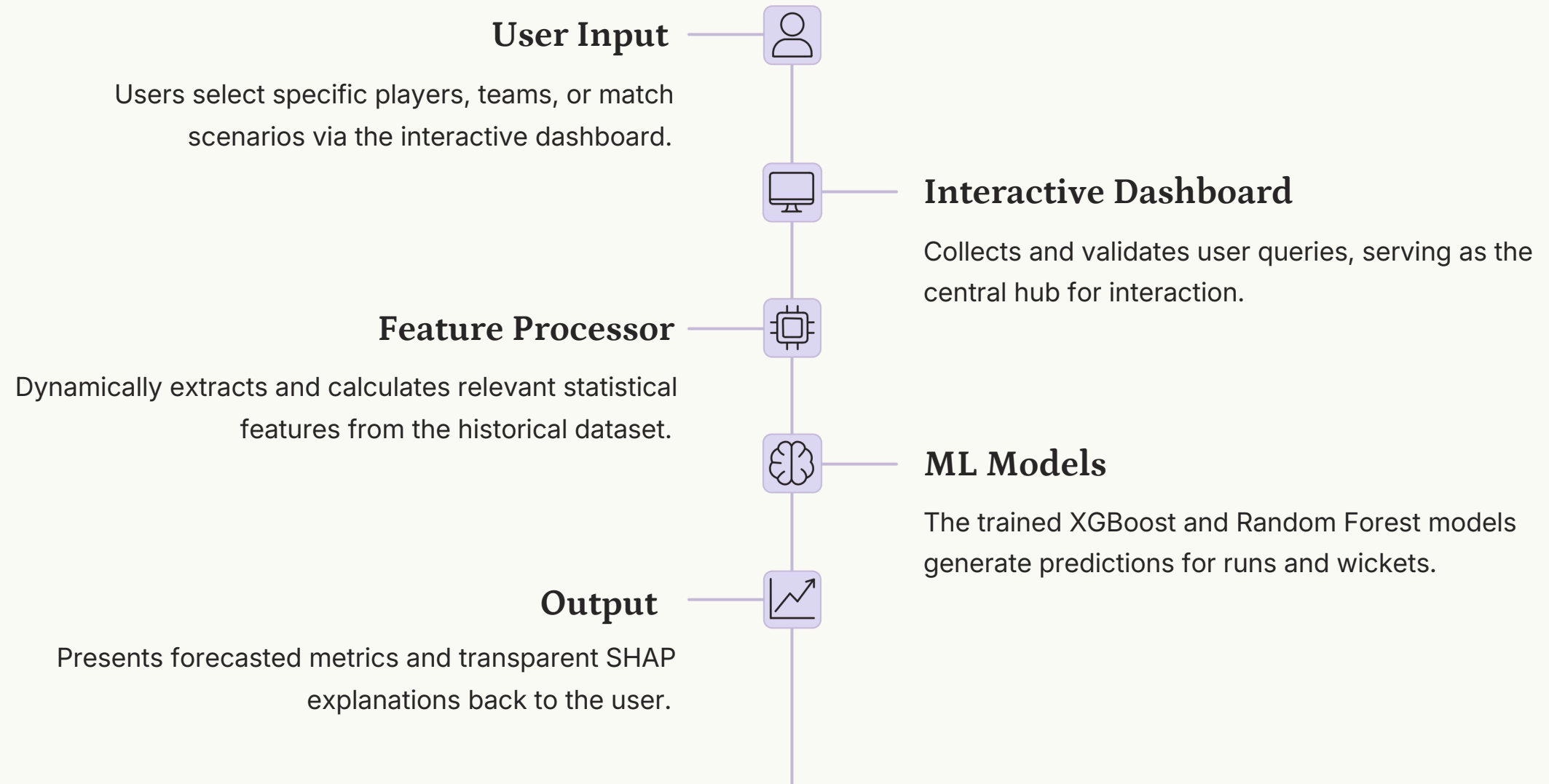


Dashboard

Url for Dashboard

Workflow

Our system streamlines the prediction process, taking user requests and transforming them into actionable insights through a robust, multi-stage workflow.



This end-to-end pipeline ensures efficient data processing and delivers timely, accurate predictions to support strategic decision-making.



Conclusion

- Add a little bitSuccessfully developed an AI-based system to predict batsman runs and bowler wickets
- Implemented complete pipeline: data cleaning, EDA, feature engineering, model training, and evaluation
- XGBoost selected for batsman prediction and Random Forest for bowler prediction based on performance
- Integrated models into an interactive Streamlit dashboard
- System provides accurate, fast, and explainable predictions using SHAP of body text



Thank You

